Hybrid finite element and neural network methods for parametric PDEs

Michel Duprez[1], Emmanuel Franck[2],
Frédérique Lecourtier[1] and Vanessa Lleras[3]

[1]Project-Team MIMESIS, Inria, Strasbourg
[2]Project-Team MACARON, Inria, Strasbourg
[3]IMAG, University of Montpellier, Montpellier

MACS seminar
("Modélisation, Analyse et Calcul Scientifique")

IMAG
INSTITUT MONTPELLIERAIN
ALEXANDER GROTHENDIECK

Context and motivation

TODO

Outline

TODO

Parametric elliptic PDEs

Let $\Omega \subset \mathbb{R}^d$ be the spatial domain ($d = 1, 2, 3$) and $\mathcal{M} \subset \mathbb{R}^p$ the parameter space.
For one or several $\boldsymbol{\mu} \in \mathcal{M}$, find $u(\,\cdot\,;\boldsymbol{\mu}) : \Omega \to \mathbb{R}$ such that

$$\begin{cases} \mathcal{L}\big(u\,;\boldsymbol{x},\boldsymbol{\mu}\big) = f(\boldsymbol{x},\boldsymbol{\mu}), & (\boldsymbol{x},\boldsymbol{\mu}) \in \Omega \times \mathcal{M}, \\ \text{Dirichlet, Neumann or Robin BC}, & (\boldsymbol{x},\boldsymbol{\mu}) \in \partial\Omega \times \mathcal{M}, \end{cases} \tag{$\mathcal{P}$}$$

with f the source term and $\mathcal{L}$ the linear differential operator defined by

$$\mathcal{L}(\,\cdot\,;\boldsymbol{x},\boldsymbol{\mu}) : u \mapsto R(\boldsymbol{x},\boldsymbol{\mu})\,u + \boldsymbol{C}(\boldsymbol{x},\boldsymbol{\mu}) \cdot \nabla u - \frac{1}{\mathrm{Pe}} \nabla \cdot \big(D(\boldsymbol{x},\boldsymbol{\mu}) \nabla u\big).$$

where R is the reaction, $\boldsymbol{C}$ the convection, D the (symmetric positive definite) diffusion matrix and Pe the Péclet number.

Numerical methods

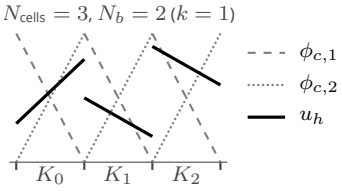
Numerical methods for parametric PDEs

Discontinuous Galerkin method[1]

Let $\mathcal{T}_h$ be a mesh of Ω of cells $(K_c)_{0 \leq c < N_{\text{cells}}}$ and $\mathcal{F}_h$ its set of faces.

Let V_h^{disc} be a discontinuous space of piecewise polynomials of degree k on the mesh $\mathcal{T}_h$. Its basis functions $(\phi_{c,i})_{c,i}$, with $\phi_{c,i}$ supported on a single cell K_c, decompose the discrete solution:

$$u_h(\boldsymbol{x}) = \sum_{c,i} \beta_{c,i}\, \phi_{c,i}(\boldsymbol{x}).$$



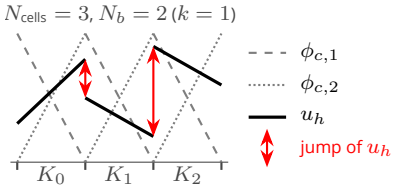
[1] [Arnold et al., 2002]

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Coupling the cells: ambiguous traces on the interfaces ⇒ numerical flux

The flux defines the method: here SIPG (Symmetric Interior Penalty Galerkin)

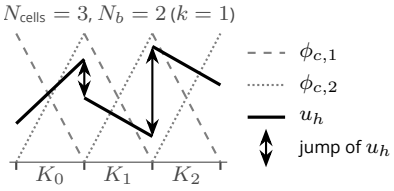
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Variational problem (SIPG): Find $u_h \in V_h^{\text{disc}}$ such that, $\forall v_h \in V_h^{\text{disc}}$, $a_h^{\text{disc}}(u_h, v_h) = l(v_h) = \int_\Omega f\, v_h$, with

$$a_h^{\text{disc}}(u_h, v_h) = \underbrace{\int_\Omega \left(R\, u_h v_h + \boldsymbol{C} \cdot \nabla_h u_h\, v_h + \frac{1}{\text{Pe}} D \nabla_h u_h \cdot \nabla_h v_h \right)}_{\text{cell terms}} \underbrace{- \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \left([\![u_h]\!] \cdot \{D \nabla_h v_h\} + \{D \nabla_h u_h\} \cdot [\![v_h]\!] \right)}_{\text{consistency}} \underbrace{+ \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \kappa\, [\![u_h]\!] \cdot [\![v_h]\!]}_{\text{penalty}}$$

where ∇_h is the broken gradient, $[\![\cdot]\!]$ and $\{\cdot\}$ the jump and average, κ the penalty parameter.

[1] [Arnold et al., 2002]

Continuous Galerkin method

TODO

Problème modèle, formulation faible
Espace $\mathbb{P}_k$ de Lagrange continu

Physics-informed neural networks

TODO

Principe, fonction de coût
Version paramétrique

1st hybridization

Enriched finite element method using PINNs

Presentation

Pipeline of the hybrid method

TODO

Additive approach

TODO

Convergence analysis

TODO

Numerical results

Test case 1
Test case 2

Numerical results

Test case 1

Test case 2

Test case 1

TODO

Numerical results

Test case 1

Test case 2

Test case 2

TODO

Conclusion, limitations and perspectives

Conclusion, limitations and perspectives

TODO

2nd hybridization

Learnable Galerkin methods

Introduction

Autodiff

TODO

Formalization

Optimization problem

TODO

Non-parametric enrichment (PINN loss)

Algorithm

TODO

Parametric enrichment (neural operator)

Algorithm

TODO

Conclusion

Conclusion

Contribution 1 : TODO

TODO

Contribution 2 : TODO

TODO

Perspectives :

- TODO

Appendix – 1st hybridization

Enriched finite element method using PINNs

Additional details

Additional details

TODO

Détails de la méthode (hors présentation principale)

Additional numerical results

Additional numerical results

TODO

Résultats numériques supplémentaires

Appendix – 2nd hybridization

Learnable Galerkin methods

Additional details

Additional details

TODO

Détails de la méthode (hors présentation principale)

Additional numerical results

Additional numerical results

TODO

Résultats numériques supplémentaires